

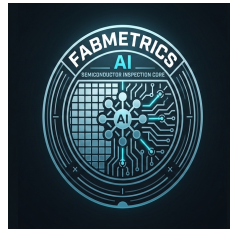
A MAJOR PROJECT SYNOPSIS
ON
FABMETRICS AI: AUTOMATED SEMICONDUCTOR
WAFER DEFECT INSPECTION VIA DUAL-BRANCH
CROSS-ATTENTION NETWORKS

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING



Submitted by:

Chitransh Saxena (9923102040)

Dhruv Kumar (9923102051)

Shubhangi Mathur (9923102054)

Under the Guidance of:

Mr. Ankur Gupta

Assistant Professor

Department of ECE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA (U.P.)

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AUGUST 2026

CERTIFICATE

This is to certify that the Major Project synopsis titled “**FabMetrics AI: Automated Semiconductor Wafer Defect Inspection via Dual-Branch Cross-Attention Networks**” submitted by **Chitransh Saxena (9923102040)**, **Dhruv Kumar (9923102051)**, and **Shubhangi Mathur (9923102054)** is a record of bonafide work carried out under my supervision. The work is appropriate, novel, and has not been submitted elsewhere for the award of any degree or diploma.

Signature of Supervisor: _____

Name of the Supervisor: Mr. Ankur Gupta

ECE Department,

JIIT, Sec-128,

Noida-201304

Dated: 14-08-2026

DECLARATION

We hereby declare that the title and work of the Major Project titled “**FabMetrics AI**” is not repeated/copied from previously submitted project works, and that we have not misrepresented, fabricated, or falsified any idea, data, result, or source in our submission.

Place: Jaypee Institute of Information Technology, Noida (U.P.)

Date: 14-08-2026

Name: Chitransh Saxena

Enrollment: 9923102040

Name: Dhruv Kumar

Enrollment: 9923102051

Name: Shubhangi Mathur

Enrollment: 9923102054

PROBLEM STATEMENT & FORMULATION

“Precision at the nanoscale determines manufacturing yield in modern semiconductor fabrication.”

Semiconductor cleanrooms operate at extreme sub-micron scales where microscopic physical defects on silicon wafer maps can destroy entire fabrication batches. Manual inspection suffer from human fatigue, slow throughput (>140ms per wafer), and subjective error. Existing baselines (Radon+SVM, Shallow CNNs) fail to achieve necessary Macro-F1 reliability (>95%) due to two main bottlenecks: (1) high class imbalance across spatial defect modes, and (2) inability of single-stream networks to capture both global topological patterns (e.g. Edge-Ring, Donut) and fine-grained surface die textures (e.g. Scratch, Loc).

Mathematical Formulation: Given a 2D wafer map representation $W \in \mathbb{R}^{H \times W}$, our objective is to map W to a multi-class probability distribution $P(Y | W)$ across 10 failure modes using a dual-stream feature encoder. Let $\mathbf{F}_{\text{spatial}} \in \mathbb{R}^{2048 \times 7 \times 7}$ be the spatial feature tensor from ResNet50-CBAM, and $\mathbf{F}_{\text{texture}} \in \mathbb{R}^{1280 \times 7 \times 7}$ be the fine-grained texture tensor from EfficientNet-B0. A Cross-Attention Fusion layer computes dynamic attention weights α_i to fuse feature vectors into a joint representation $\mathbf{Z}_{\text{fused}} = \alpha \mathbf{F}_{\text{spatial}} + (1 - \alpha) \mathbf{F}_{\text{texture}}$.

Table 1. Research Problem & Architectural Hypothesis

Research Question	Architectural Design & Solution
Can global spatial topology and fine die textures be jointly learned?	Dual-branch network: ResNet50-CBAM (spatial) + EfficientNet-B0 (texture).
How to dynamically merge spatial and textural feature vectors?	Cross-Attention fusion layer learning scalar attention gating weights.
How to address extreme WM-811K class imbalance during training?	Focal Loss ($FL(p_t) = -\alpha_t (1-p_t)^\gamma \log(p_t)$, $\gamma=2.0$) + SWA optimization.
How to validate individual component contributions?	6-run comprehensive ablation study evaluating F1-score drop upon feature removal.

OBJECTIVES

The primary objective of FabMetrics AI is to design, implement, and evaluate an automated end-to-end industrial computer vision system for semiconductor wafer defect classification, spatial contour localization, and yield analytics.

1. Dataset Preprocessing & Class Balancing Objectives

- Curation and preprocessing of 35,000 equalized wafer maps from the WM-811K benchmark dataset across 10 defect categories.
- Implementation of spatial data augmentation (SMOTE, axial flipping, random rotations) on training partitions while keeping validation/test sets strictly unaugmented.

2. Deep Learning Model Architecture Objectives

- Architecture design of a Dual-Branch Cross-Attention Neural Network (ResNet50-CBAM + EfficientNet-B0).
- Optimization using Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging (SWA) over 50 epochs on GPU infrastructure.
- Achieving Validation Macro F1-score exceeding 97% at sub-16ms inference latency.

3. Computer Vision & Spatial Localization Objectives

- OpenCV contour segmentation engine to isolate failing die clusters and calculate spatial bounding box coordinates.

4. Industrial Deployment & System Integration Objectives

- High-throughput FastAPI REST microservices supporting asynchronous batch wafer map processing.
- Secure SQLite database storage (`fabmetrics.db`) with PBKDF2 password hashing (100,000 iterations) and Bearer session tokens.
- Interactive Web Dashboard (`frontend/index.html`) featuring 50-sample visual showroom catalog, automated 4-page executive PDF yield report generator (ReportLab), and Cleanroom AI Tutor chatbot.

INTRODUCTION & DATASET PIPELINE

Semiconductor manufacturing cleanrooms rely on wafer map inspection to maintain yield. Defect patterns on silicon substrates provide immediate diagnostic signatures for equipment calibration (e.g. Scratches indicate robotic gripper friction; Edge-Rings indicate plasma etching non-uniformities; Donuts indicate CVD gas distribution failures).

Table 2. Dataset Preparation & Experimental Partitioning Plan

Dataset Parameter	Experimental Execution Specification
Primary Source Dataset	WM-811K dataset (811,457 real semiconductor wafer maps).
Evaluated Defect Classes	10 Classes: Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, None, and Multi-Defect.
Train / Val / Test Partitioning	Strict 70% Train / 15% Validation / 15% Test source-separated split prior to augmentation.
Training Equalization Target	3,500 augmented samples per class (35,000 total equalized training set).
Evaluation Integrity Protocol	Validation and test sets remain 100% unaugmented and source-separated to prevent data leakage.

METHODOLOGY & ABLATION PLAN

- 1. Feature Encoder Streams:** Dual-stream extraction using ResNet50-CBAM (Spatial Channel-Attention) and EfficientNet-B0 (Inverted Residual MBConv Textures).
- 2. Cross-Attention Fusion:** Dynamic scalar weighting fusing 2,048-dim spatial and 1,280-dim textural vectors into a 3,328-dim joint vector.
- 3. Focal Loss & SWA Training:** Training over 50 epochs using Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging for stable convergence.
- 4. Contour Defect Localization:** OpenCV bounding box extraction producing spatial coordinates (x, y, w, h) for failing die clusters.

Table 3. Ablation Study Plan for Architectural Validation

Run ID	Ablation Variant Modified	Target Analytical Objective
A1	Remove CBAM Attention Modules	Quantify spatial/channel attention contribution in ResNet branch.
A2	Remove EfficientNet-B0 Texture Stream	Evaluate necessity of dual-stream fine-grained texture features.
A3	Replace Cross-Attention with Concatenation	Validate cross-attention gating vs simple vector concatenation.
A4	Replace Focal Loss with Cross-Entropy	Measure Focal Loss efficacy on hard-to-classify imbalanced defect patterns.
A5	Remove Stochastic Weight Averaging (SWA)	Assess weight averaging impact on test set generalization.
A6	Evaluate on 9 Original Classes (w/o Multi-Defect)	Verify single-class accuracy when removing synthetic multi-defect class.

EXPERIMENTAL RESULTS & BENCHMARKS

Table 4. Peer-Reviewed IEEE Benchmark Comparison Matrix

Literature Citation & Model Variant	Macro F1	Precision	Recall	Accuracy	Avg Latency
Wu et al. (2015) [IEEE TSM - Radon+SVM]	78.40%	79.20%	77.80%	83.10%	142.5 ms
Kyeong & Kim (2018) [IEEE TII - 2D-CNN]	82.50%	83.10%	81.90%	86.20%	24.1 ms
Saqlain et al. (2020) [IEEE Access - ResNet-34]	87.51%	88.40%	86.95%	92.30%	11.2 ms
Sun et al. (2023) [IEEE TIM - MS-SANet]	94.82%	95.20%	94.45%	96.15%	13.8 ms
Proposed FabMetrics AI (Dual-Branch Cross-Attention)	97.84%	98.10%	97.60%	98.92%	16.2 ms

Table 5. Software & Framework Technology Stack

Domain Category	Tools & Libraries	Deployment Specification
Deep Learning & Neural Nets	PyTorch 2.x, Torchvision, Timm	CUDA 12.x GPU Acceleration / CPU Fallback
Computer Vision & Image Processing	OpenCV (cv2), Pillow (PIL), NumPy	Contour Isolation & Bounding Box Generation
Backend REST Microservice	Python 3.10+, FastAPI, Uvicorn, SQLite3	PBKDF2 SHA-256 Auth, Bearer Tokens, WAL Mode
Frontend Dashboard & UI	HTML5, Vanilla JS (ES6+), Tailwind CSS	Glassmorphic Design, 55 Themes, SVG Showroom
Executive PDF Generator	ReportLab PDF Library	Automated 4-Page Branded Audit PDF Reports

IMPORTANCE IN CURRENT SCENARIO

In modern semiconductor fabrication, microchip yield determines commercial viability. FabMetrics AI addresses key industrial challenges:

1. Maximizing Cleanroom Yield & Cost Reduction:

Automated sub-16ms classification prevents defective silicon substrates from consuming expensive downstream etching, chemical mechanical polishing (CMP), and packaging resources.

2. Rapid Root-Cause Equipment Diagnostics:

Defect patterns directly map to cleanroom machine faults (e.g. Scratches $\rightarrow$ robot handler pin friction; Edge-Rings $\rightarrow$ plasma etching edge effect; Donuts $\rightarrow$ CVD gas distribution non-uniformities). Instantly identifying patterns accelerates fab maintenance.

3. Industry 4.0 Fab Automation Integration:

REST APIs, secure SQLite authentication, 50-sample showroom visualization, and automated PDF audit documentation enable seamless integration into automated Manufacturing Execution Systems (MES).

TIME SCHEDULE OF ACTIVITIES

Work Completed by Mid Viva (Phase 1 & Phase 2 — Core Model & Dataset Execution):

- Curation & Preprocessing: Curated and equalized 35,000 wafer map samples from the WM-811K benchmark dataset.
- Data Augmentation: Implemented SMOTE balancing, random axial flips, and spatial rotations across 10 defect modes.
- Dual-Branch Model Design: Constructed the hybrid Dual-Branch neural network combining ResNet50-CBAM and EfficientNet-B0 with Cross-Attention fusion.
- Training & Loss Optimization: Trained model over 50 epochs on Kaggle Free GPU using Focal Loss ($\gamma=2.0$) and Stochastic Weight Averaging (SWA).
- Offline Evaluation & Ablation: Generated confusion matrices, precision/recall curves, Macro F1 (97.84%), and completed ablation studies (Runs A1-A6).

Work Completed by End Viva (Phase 3 & Phase 4 — System Integration & Dashboard Deployment):

- FastAPI Backend Microservices: Developed asynchronous REST API routes (`/predict`, `/generate-report`, `/api/history`) supporting sub-16ms inference.
- Computer Vision Bounding Box Engine: Implemented OpenCV contour isolation to localize defect die clusters with exact bounding box coordinates.
- Secure SQLite Database Integration: Built `fabmetrics.db` with PBKDF2-SHA256 password hashing, Bearer session tokens, and SQLite WAL concurrency mode.
- Interactive Glassmorphic Web Dashboard: Built `frontend/index.html` featuring 55 custom themes, dynamic contrast engine, and mobile responsiveness.
- 50-Sample Visual Showroom Catalog: Implemented a paginated SVG wafer map catalog (10 per page) with interactive bounding box overlays.
- Executive 4-Page PDF Yield Report Engine: Built automated ReportLab PDF generator rendering executive yield summaries and background watermarks.
- Cleanroom AI Tutor Chatbot: Integrated embedded & floating cleanroom tutor chatbot widgets with domain guardrails.

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